

MUHAMMAD KAZIM, Ph.D.

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Interdisciplinary AI Researcher | Energy Systems, IoT & Research Computing

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RESEARCH PROFILE

Interdisciplinary AI researcher and educator with a Ph.D. in Industrial and Manufacturing Engineering from North Dakota State University, United States of America. My research develops graph-based and machine-learning methods for resilient energy infrastructure, low-power IoT networks, sustainable wireless systems, cybersecurity, and biomedical discovery. I combine network science with domain-specific simulation and research computing. My work includes peer-reviewed publications, funded NSF and USDA research, and collaborations using utility-scale operational data. I currently serve as Visiting Faculty in the Artificial Intelligence Department at the University of Malakand.

EDUCATION

- **Doctor of Philosophy (Ph.D.), Industrial & Manufacturing Engineering, North Dakota State University, Fargo, USA, Conferred August 2025.**
 - **Thesis:** Enhancing Resilience in Energy Networks: A Multilayer Network and Machine Learning Approach
 - **Project:** AISustein Project (AI for Resilient Energy Infrastructure) aisustein.com/research
 - **Focus Areas:** Multilayer Networks, Graph Neural Networks, Energy System Vulnerability, Predictive Maintenance.
- **Master of Science (MS), Computer Science, Abasyn University, Peshawar, Pakistan, Conferred August 2019**
 - **Thesis:** Enhanced Detection Mechanism for Application-Layer DDoS Attacks using Machine Learning.
- **Master of Science (M.Sc), Computer Science, University of Malakand, KPK, Pakistan, Conferred March 2014**
 - **Awarded Gold Medal** for securing First Division with distinction.
 - **Focus:** Artificial Intelligence, Software Engineering, and Data Networks.

ACADEMIC & PROFESSIONAL EXPERIENCE

I. Visiting Faculty Dept. of Artificial Intelligence, University of Malakand, Pakistan / September 2025 – Present - Teaching undergraduate and graduate courses in AI, Data Structures, and Programming. - Supervising research projects in intelligent systems and machine learning. - Contributing to departmental development under the Faculty of Computing & Engineering.

II. Graduate Research Assistant, Dept. of Industrial & Manufacturing Engineering, NDSU / June 2023 - August 2025 (AISustein collaboration) - Developed multilayer-network and GNN models for energy resilience, predictive maintenance, and fault propagation using Belgian power-infrastructure data and seven years of OGE incident records. - Used SAInt, MATLAB, and NDSU CCAST HPC for energy-system simulation and large-scale model training.

III. Graduate Teaching Assistant, Dept. of Computer Science, NDSU / Aug 2022 – May 2023
- Led lab sessions for CSCI 172 (Intermediate Visual Basic). - Evaluated student projects and provided technical mentorship.

IV. Data Processing Supervisor (C/O) BISE Malakand, Pakistan / June 2017 – April 2022 -
Automated transcript processing & payroll for 120+ employees - Deployed Oracle PL/SQL & OMR for student evaluation

V. Software Engineering & Systems Development (2014–2017) LGE&RDD, Pakistan / June 2014 – 2017 - Created government platforms for citizen services & e-bidding systems. - Upgraded web apps using Core PHP, CodeIgniter, Oracle, and AngularJS

INTERDISCIPLINARY RESEARCH EXPERTISE

- **AI for Sustainable Energy and Critical Infrastructure:** AI-driven modeling of smart grids and interconnected energy systems, including predictive maintenance, infrastructure resilience, vulnerability assessment, cascading-failure analysis, and sustainable energy optimization.
- **Network Science and Graph Learning:** Multilayer and multiplex networks, complex systems modeling, graph neural networks, link prediction, network navigability, interdependency analysis, graph embeddings, and resilience evaluation.
- **Machine Learning and Explainable AI:** Deep learning, graph representation learning, predictive modeling, edge-level explainability, benchmarking of machine-learning models, autonomous systems, and decision-support frameworks.
- **Internet of Things and Network Security:** Security of RPL-based low-power and lossy IoT networks, multilayer attack modeling, routing and control-plane vulnerability analysis, network-managed intrusion detection, and computational-overhead assessment.
- **Sustainable Wireless and 5G/B5G Networks:** Energy-efficient radio access networks, user association, base-station sleep management, switching-aware graph control, quality-of-service-aware optimization, and sustainable telecommunications infrastructure.
- **Cybersecurity and Digital Forensics:** Deepfake detection, digital media authenticity, privacy protection, browser artifact analysis, anonymity assessment, and intelligent cyber-threat detection.
- **Biomedical AI and Computational Biology:** Network-integrated reverse vaccinology, protein-protein interaction analysis, AI-assisted vaccine-candidate discovery, Biomni-based feature prioritization, and graph learning for biological systems.

RESEARCH COMPUTING AND SIMULATION CAPABILITIES

- **Integrated Energy Systems Simulation:** SAInt for electricity, gas, and coupled gas-electric network modeling, scenario analysis, infrastructure resilience, and operational studies.
- **Cyber-Physical & IoT Simulation:** Cooja/Contiki-NG for RPL-based low-power and lossy network experiments, multi-attack scenarios, routing/control-plane analysis, and communication-overhead evaluation.
- **High-Performance Research Computing:** NDSU CCAST for large-scale scientific computation, graph-model training, simulation, and reproducible computational workflows.

- **Engineering Computing and Modeling:** MATLAB for numerical analysis, algorithm prototyping, energy-system modeling, simulation, and data analysis.
- **Machine Learning and Graph Analytics:** Python, PyTorch, TensorFlow, scikit-learn, GCN, GAT, GraphSAGE, NetworkX, Pandas, NumPy, and explainable AI.
- **Research Software and Data Infrastructure:** C++, R, Linux, Git, Docker, Jupyter, LaTeX, SQL, Oracle, MySQL, PostgreSQL, and scientific visualization.

FUNDED RESEARCH

1. AISustain: AI for Resilient Energy Infrastructure - Funding Agency: National Science Foundation (NSF) - **Project Website:** <https://www.aisustain.com/about> - **Role:** Graduate Research Assistant / Ph.D. Researcher - **Period:** June 2023 – August 2025 - **Description:** Primary funding source for doctoral research on multilayer energy networks and predictive maintenance. This project provided access to real-world utility data from Oklahoma Gas & Electric Company (OGE) spanning seven years of incident reports, as well as the Belgian power infrastructure dataset. Research outcomes include four peer-reviewed journal publications and one conference paper on GNN-based predictive maintenance, link prediction, and network resilience.

2. Network-Integrated Reverse Vaccinology for Flavobacterium - Funding Agency: U.S. Department of Agriculture (USDA) - **Role:** Graduate Research Assistant - **Period:** Summer 2025 - **Description:** Developed a novel computational framework integrating a biomedical AI agent (Biomni) with Graph Neural Networks (GNNs) to identify vaccine candidates for *Flavobacterium*, a significant pathogen in aquaculture. This work resulted in a preprint and is currently under peer review.

INDUSTRY AND UTILITY PARTNERSHIPS

- **Oklahoma Gas & Electric Company (OGE):** Collaborated on a comprehensive analysis of seven years of real-world incident data from a major North American utility. This partnership resulted in the development and validation of predictive maintenance models that demonstrated commercial viability and real-world applicability of GNN-based approaches for grid resilience.
- **Research Collaboration:** Demonstrated ability to work with industry partners to access proprietary operational data and translate academic research into practical solutions for critical infrastructure management.
- **Commercialization Pathway:** Committed to translating research into industry-ready solutions through partnerships with utility operators and energy sector stakeholders.

RESEARCH PROPOSALS IN PREPARATION

- **Title:** Self-Healing Energy Grids with Autonomous AI Agents
Concept: To develop an autonomous grid management system that integrates a Semantic Energy Knowledge Graph (SEKG), Multilayer GNNs, and autonomous AI agents to predict and prevent cascading failures, and optimize energy flow in real-time.
- **Title:** A Scalable AI-Powered Platform for Rapid Vaccine Discovery
Concept: To create a generalizable, modular platform for rapid vaccine candidate discovery across multiple pathogen systems by integrating multi-omics data with advanced GNN architectures.

RESEARCH PUBLICATIONS

Journal Articles

1. **Kazim, M.**, Pirim, H., Shi, S., & Wu, D. (2024). Multilayer analysis of energy networks. *Sustainable Energy, Grids and Networks*, 39, 101407. <https://doi.org/10.1016/j.segan.2024.101407>.
2. **Kazim, M.**, Pirim, H., Yadav, O. P., Le, C., & Le, T. (2025). Analysis of multilayer energy networks: A comprehensive literature review. *Applied Energy*, 398, 126357. <https://doi.org/10.1016/j.apenergy.2025.126357>.
3. **Kazim, M.**, Pirim, H., Le, C., Le, T., & Yadav, O. P. (2025). Multilayer GNN for Predictive Maintenance and Clustering in Power Grids. *iScience*.
4. **Kazim, M.**, Pirim, H., Yadav, O. P., Le, C., & Le, T. (2025). Edge-level explainable graph neural networks with network-centric features for transmission line failure prediction in power grids. *Sustainable Energy, Grids and Networks*. <https://doi.org/10.1016/j.segan.2025.101969>.
5. Sohail, S., Sajjad, S. M., Zafar, A., Iqbal, Z., Muhammad, Z., & **Kazim, M.** (2025). Deepfake Image Forensics for Privacy Protection and Authenticity Using Deep Learning. *Information*, 16(4), 270. <https://doi.org/10.3390/info16040270>.
6. Javed, M. S., Sajjad, S. M., Mehmood, D., Mansoor, K., Iqbal, Z., & **Kazim, M.** (2024). Analyzing Tor Browser Artifacts for Enhanced Web Forensics, Anonymity, Cybersecurity, and Privacy in Windows-Based Systems. *Information*, 15(8), 495. <https://doi.org/10.3390/info15080495>

Conference Proceedings

7. **Kazim, M.**, Pirim, H. et al., Link Prediction and Navigability of Multiplex Energy Networks, *ASCE International Conference on Computing in Civil Engineering*, July 28-31, 2024

Under Review

8. **Kazim, M.**, Farzana, F., Pirim, H., Hanson, L., Griffin, M., & et al. (2025). Network-Integrated Reverse Vaccinology Using Biomni-Prioritized Features and Graph Neural Networks in *Flavobacterium*. [Preprint](#).
9. Abbas, S., **Kazim, M.**, Fayaz, M., and Pirim, H. (2026). Overhead-Aware Multilayer Graph Learning for Network-Managed Multi-Attack Detection in RPL-Based Low-Power IoT Networks. Preprint under review. [Preprint](#)
10. **Kazim, M.**, and Pirim, H. (2026). From Prediction to Action: A Controlled Benchmark of Multilayer Graph Embeddings for Substation-Level Predictive Maintenance. Preprint under review. [Preprint](#)
11. **Kazim, M.**, Abbas, S., and Fayaz, M. (2026). Switching-Aware Multilayer Graph Control for Sustainable 5G/B5G RANs: Energy-Efficient User Association and Base-Station Sleep Management. SSRN preprint. [Preprint](#)